EDO STATE COLLEGE OF AGRICULTURE

SCIENCE AND TECHNOLOGY, IGUORIAKHI,

NIGERIA

DEPARTMENT OF AGRICULTURAL

TECHNOLOGY

COURSE CODE: AGT 126

COURSE TITLE: MICROLIVESTOCK

PRODUCTION

COURSE UNIT: 4

GENERAL OBJECTIVES

- Know the important species of animals regarded as micro-livestock species.
- Know the types of housing and equipment required for micro-livestock production.
- Understand the nutrition and feeding of micro-livestock.
- Understand reproduction and breeding of micro-livestock.
- Understand the routine management for different species of micro-livestock.
- Know the common diseases of micro-livestock animals and their control.

What are Micro-Livestock species?

Microlivestock is often defined as species that are inherently small, as well as breeds of animals that are smaller than the average breed. Micro livestock can also be referred to as a class of farm animals that are reared in small scale. There are two types of microlivestock. One consists of extremely small forms of conventional livestock—such as cattle, sheep, goats, and pigs. The other consists of species that are inherently small—poultry, rabbits, and rodents, for instance.

For example, most poultry animals are classified as micro livestock--chickens, ducks, turkeys. Even miniature breeds of goats, pigs, and cattle are classified within the term. In communities where space or resources are limited, microlivestock provide excellent opportunity for the population to sustain themselves. For example, The “mini-Brahman” cow of Mexico is only 60 cm tall and weighs 140 kg; the southern Sudan dwarf sheep of eastern Africa can weigh as little as 11 kg; the Terai goat of Nepal weighs less than 12 kg; and the cuino pig of Mexico weighs merely 10 kg.

Poultry breeds which could be considered as microlivestock include: quail, duck, guinea fowl, and turkey.

Species

- Rabbit
- Edible land snail
- Grass cutter/cane rat
- Giant bush rat
- Guinea pigs
- Pigeons
- Quails

Benefits of rearing microlivestock

Economic

Microlivestock lend themselves to economic niches that are not easily filled by large livestock. Much of their potential is for subsistence production. They are promising for the many peasants who, being outside the cash economy, are now unable to purchase meat, milk, cheese, or eggs. These people can afford only animals that can be raised within the home or backyard under ambient climatic conditions and on feeds that are cheap and easily available.

A subsistence farmer is likely to benefit more from small species than from large because of several factors:

- The animals are less expensive to buy.
- They are less of a financial risk to maintain. (A farmer with several small animals is less vulnerable to loss than a farmer with a single large animal, a feature that is particularly important in subsistence farming where success determines whether the family will survive.)
- They give a faster return on investment. (Small size generally signifies high reproductive capacity and a fast turnover.)
- They provide flexibility. (Farmers can more easily change the size of their herd or flock to match the amount of feed available at a given time. Also, they can sell animals according to the family's fluctuating needs for cash or food.)
- They provide a steadier source of income.
- They increase the chances of successful breeding because greater numbers are usually kept. (This also means that breeding stock is more likely to be retained in times of scarcity.)

- They are more easily transported.
- In some cases they are more efficient converters of food energy.

Other benefits include:

- Reduced Spoilage. A portion of meat that comes in a “package” of a size that can be readily consumed by a family is important in areas where refrigeration is unavailable or uneconomic. A family can eat the meat produced by most microlivestock in one meal or in one day to minimize the risk of spoilage.
- Efficient use of space. The space required for handling and feeding microlivestock is proportionately less than that required for large animals. Low space requirements make many microlivestock (such as guinea pigs, rabbits, pigeons, and quail) available to landless rural inhabitants who have no room for a cow. This is particularly important with respect to feed production.
- Cheaper Facilities. Facilities and equipment required for microlivestock are, by and large, smaller and simpler than those required for large animals. They often can be made from local products or scrap material or both.
- Ease of Management. Farmers and villagers can manage small animals more easily than large, which is an advantage in the many places where women and children are the main keepers of livestock.
- Increased Productivity. Small animals tend to fit well into existing farming systems, thereby expanding the resource base and recycling nutrients. Some— for example bees, ducks, and geese—can feed themselves by scavenging.
- By-Products. Many species have fur, feathers, skins, and other by-products that are often more valuable than their meat, milk, or eggs. Examples include the feet and tails of rabbits, musk from musk deer, and pelts from rodents such as coypu. Processing such by-products creates diversification for the farmer and perhaps jobs for the village.

Feed

In general, small species tend to expand the food base by using a wider array of resources than do major livestock such as cattle. Many can be raised on feeds that people discard: fibrous residues, industry by-products, or kitchen wastes. Some collect minute feeds that otherwise go unused. For example, chickens and pigeons gather scattered seeds, turkeys gobble up insects, geese graze water weeds, iguanas feed in the tops of trees, and bees collect nectar and pollen from flowers that may be miles away.

Even some grazing microlivestock prefer different forages from those preferred by cattle. Antelope and deer, for instance, browse tree leaves; capybara and grasscutters eat reeds. Combining microlivestock with conventional livestock results in a more complete utilization of forage resources and greater animal production per hectare.

Under conditions of abundance, small size may be of no advantage in mammals, but if feed is limited, it is of great help. A small animal (or its keeper) needs to cover less area to fulfill its daily requirements, so that microlivestock may grow fat in areas where the forage is too sparse to support a larger animal. This is particularly vital when there are seasonal bottlenecks. For example, feed may be plentiful enough for most of the year to supply many large animals; however, the dry season may greatly restrict the numbers that can be kept.

Although small animals generally require proportionately higher inputs of feed, they also grow proportionately faster. In addition, species such as rabbits, guinea pigs, and grasscutters digest fibrous matter with surprising efficiency, even though they are not true ruminants like cattle, sheep, and goats.

Reproduction

Many small animals have high reproductive capacity with short gestation periods, large numbers of offspring, and rapid juvenile growth. They tend to reach sexual maturity at a younger age than large animals, and the interval between the generations can be very short. Thus, meat or other products can be produced more rapidly and more evenly throughout the year.

Cows, for example, produce a maximum of one calf per year. A pig, on the other hand, may produce 7 or more young; a rabbit, 30 or more; a chicken, more than 100.

Adaptability and Hardiness

The survival rates and manageability of many small breeds and species can be outstanding. Smallness is often an adaptation to harsh environment. Indeed, a major promise for microlivestock is in special environmental niches. Where cold, heat, temperature fluctuations, aridity, or humidity are extreme, microlivestock are likely to show their greatest advantage. Chickens, guinea fowl, goats, and many other small species already live in villages, homes, and backyards in harsh and disease-prone climates, and are usually given no care and sometimes no food: they have to scavenge for their sustenance and survive as best they can. Such selection pressures result in animals of remarkable adaptability, tolerance, and robustness.

Some microlivestock can produce under conditions where conventional species die. The capybara, for instance, is at home in the Latin American lowlands, where the climate is hot and humid and floods cause seasonal inundations. Cattle, by contrast, die because of malnutrition, foot rot, or drowning. Other microlivestock species with a wide tolerance to ecological extremes include the turkey, pigeon, and bee. And some dwarf breeds of cattle, sheep, goats, and pigs show surprising tolerance to trypanosomes, the parasites that make conventional breeds impossible to maintain throughout much of Africa.

Some small species can be raised in cities, where poverty and malnutrition are often worse than in rural areas. It is estimated, for instance, that one million livestock exist in Cairo, not counting the pigeons that are raised on countless rooftops. Goats and cattle are common in urban India, and many Third World cities have far more chickens than people

DISTRIBUTION OF MICRO-LIVESTOCK SPECIES

Rabbit

The wild ancestor of the domestic rabbit was originally restricted to Spain and Portugal. Today, its descendants are found worldwide. Domestic rabbits are best suited to temperate climates, but they do well in tropical and subtropical conditions if hutches are constructed and sited to take advantage of shade and cooling breezes. Ventilation is important (but care must be taken to avoid direct exposure to cold drafts). Prolonged exposure to temperatures higher than 30°C reduces both fertility and growth. Apparently, all breeds tolerate heat equally well. However, heat is shed through the ears, and the longer the ear, the more heat a rabbit will tolerate. Lop-eared varieties withstand heat poorly.

Grass cutter

Grasscutters occur in grassland or in wooded savanna throughout the humid and subhumid areas of Africa south of the Sahara. They often live in forest-savanna habitats where grass is present. They do not inhabit rainforest, dry scrub, or desert, but they have colonized the road borders in forest regions. Distribution is determined by availability of adequate or preferred grass species for food. Specifically, *Thryonomys swinderianus* occurs in virtually all countries of west, east, and southern Africa. *Thryonomys gregorianus* occurs in savannas in Cameroon, Central African Republic, Zaire, Sudan, Ethiopia, Kenya, Uganda, Tanzania, Malawi, Zambia, Zimbabwe, and Mozambique.

The larger grasscutter (*T. swinderianus*) generally lives in swampy, low-lying areas, especially along river banks and the borders of lakes and streams. Occasionally, it is found on higher ground among bushes and rocks, living where savanna grasses are dense and tangled enough to afford good cover. In Ivory Coast and southern Guinea, for instance, grasscutters are found (and hunted) throughout the savanna zones. And they can occur in close proximity to farmlands and people (for example, in southwest Nigeria).

Guinea pigs

The original home of the wild guinea pig is believed to have been the central highlands of Peru and Bolivia. Its domesticated descendants are important as meat animals mainly in that same area, but, as noted, they are also important in certain African and Asian countries. A few strains are distributed worldwide as laboratory animals and pets.

These extremely adaptable animals are found in temperate zones and in the highland tropics, but they are usually kept indoors and protected from the extremes of weather. In Lambayeque and other departments of Peru, they are reared at elevations from sea level to more than 4,000 m. In areas where they are raised, daily temperatures fluctuate as much as 30°C. In the Bolivian or Peruvian puna region, for instance, day temperatures can be 22°C, while night temperatures are –7°C. However, they cannot survive freezing temperatures and they may not perform well when exposed to the full tropical heat and sunlight. Many people of the Peruvian highlands keep the animals in darkness (for example, in wood boxes with little or no light). The animal's original wild habitat is believed to have been an area of grasslands, forest edges, swamps, and rocks.

Quails

The ancestral wild species is widely distributed over much of Europe and Asia as well as parts of North Africa. Although domestic quail are now available almost everywhere, Japan is probably the world leader in commercial production; quail farms are common throughout its central and southern regions.
Quail are hardy birds that, within reasonable limits, can adapt to many different environments. However, they prefer temperate climates; the northern limit of their winter habitat is around 38°N.

Pigeons

The wild ancestor of the common pigeon—domestic, wild, or feral—is thought to be the rock pigeon or rock dove of Europe and Asia. Today its domestic descendants are bred in virtually every country, and those that have gone feral (reverted to the wild) occur in most of the world's cities and towns.

The domestic pigeon can be raised equally well in temperate and tropical zones. Indeed, this adaptable species can be kept anywhere that wild pigeons exist, including arid and humid regions. It should be noted, however, that cold climates do not favor squab production and hot climates promote vermin and disease.

Giant rat

Giant rats are commonly found from Senegal to Sudan, and as far south as the northern region of South Africa. The main species is mostly found in moist savannas, patches of forests, and rainforests. However, it can also be found in all West African vegetation zones from the semiarid Sahel to the coast. It also exists at high altitudes—up to about 2,000 m in West Africa and 3,000 m in eastern Africa.

The rainforest species occurs in the great equatorial forest belts of Zaire and neighboring Central African countries.

Giant rats occur largely in lightly wooded dryland regions or in forested humid regions. They cannot tolerate high temperatures or truly arid conditions. They often live in farm areas and in gardens. Their burrows are commonly found inside deserted termite mounds and at the base of trees. Some have also been found in the middle of cassava fields.

HOUSING AND EQUIPMENT

Housing provides a micro-climate for livestock, and allows careers to modify the environment for a better living condition. Different livestock species have specifications that best suit them. However, livestock that falls with close category usually have a similar behavioral and physiological needs, and therefore similar housing requirements. The above listed micro-livestock species can be categorized into: Poultry (quail, guinea fowl, pigeon), pseudo ruminants (rabbit, guinea pigs, giant rat, grass cutters).

Housing and equipment for pseudo ruminant production

There is no one style fits all in the case of rabbit housing that is suitable for all situations, but there are certain basic requirements to be met regardless of the type of housing used. These requirements include:

- Comfort
- Confine rabbit from escape
- Protect rabbit from predators and adverse weather
- Allow easy, comfortable access for careers
- Self-cleaning or easy to clean
- Cost of housing should be reasonable

Types of housing

Housing rabbit can be either outdoors or within a building. Outdoor housing is often less expensive than indoor housing and because of better ventilation, the rabbit are usually healthier in outdoor units. However, security is more difficult to provide with outdoor units, and the protection from the weather may not be as good as indoors. This protection applies to the manager as well as the rabbits. In severe weather, rabbit houses where the manager is exposed to the element of weather may not get the care they need because their manager is uncomfortable and in a hurry. It is also more difficult to provide controlled lighting if the rabbits are housed outdoors. Outdoor lighting helps provide security for rabbitries that are not inside a building.

1. Indoor housing are more expensive than outdoor housing but provide better weather protection better security and more management efficiency than the outdoor units. The disadvantages of the indoor housing include the cost problem with temperature and humidity control and difficulty with ventilation. There may be a greater incidence of disease, especially respiratory problems. Indoor housing generally involves the use of cages made entirely of wire. These are also usually the most expensive and must be used indoors or in a shelter.
2. Free-standing outdoor cages (hutches) can be made of a variety of materials. Wire cages with plastic or sheet metal tops, backs and ends will provide good protection from the weather but are expensive and also will reduce ventilation to the rabbits. Whenever sheet metal or plastic is used for roofs, insulation must be placed between the roof and the rabbits. Sunlight beating down on the roof will cause heating and if no insulation is provided, this heat will be radiated onto the rabbit which could make them uncomfortable.

The most common type of outdoor hutch is a combination of wire and wood. The wood is used to make a frame, the wire for the floor and some of the sides or ends, and sheet metal or other material provides a waterproof roof. Wood included in rabbit housing can cause problems. Rabbits are gnawers and one of the things they like to chew is wood. As a result, wood is used in rabbit housing should be protected to keep the rabbits from gnawing on it and should pressure-treated to retard rotting.

House orientation

In constructing a rabbitry, an east-west orientation is preferred. To provide good air circulation, the width of rabbit house must not exceed 8 meters (m). Window space must represent not less than 25% from rabbitry floor space. The roof should be 3.2 to 3.5m high, with slopes south-north to avoid exposure to vertical heat of the sun. The top and outer walls of the building should be painted white to reflect heat as much as possible. Planting shade trees around the rabbitry helps to cool the rabbitary in hot weather as well as reduced drafts.

Cage type

Wire cages may be either the rectangular or gouset style. The rectangular cages are further divided into the front-opening or top opening. The front opening rectangular cage is useful in situations where the cages are to be stacked or if the doors swing inward, where aisle space is limited. In that case, the door should be placed on one side of the front of the cage and the feeder on the other. This may make it difficult to reach into the far corner of the cage.

Hutches

Rabbit hutch is a free standalone unit in which one or several rabbits can be housed. Hutches usually have their own protection from the weather and other predators. In constructing a hutch, some provision has to be made for legs or supports, for the cage area where the rabbit is to be used for the roof and sides which will provide a protection from weather for the rabbits. Most outdoor hutches are supported on wooden legs, either poles or treated 4 x 4 timbers that the rabbits off the ground, provide space for the buildup of a manure pack, elevate the rabbits so they will be convenient to manage and to keep them from being harassed by dogs, cats and other neighborhood pest. The legs should be treated against termite and rotting if they are made of wood.

Selecting a foundational stock

In the selection of foundational stock, there are 2 key aspects to assess. They are:

A. Visual assessment
    a. Check the entire rabbitry for cleanliness, contented calm stock and an abundance of plump young rabbits.
    b. Notice the number of young in each litter (2/3 of the working does should have litters)
    c. There should not be strong ammonia fumes or rabbit sneezing in the hutches.
    d. Check the coat is in good condition. Reject wooly, sparse or dull fur.
    e. The eyes should not be weepy but clear and bright
    f. Teeth should not be yellow or deformed
    g. Check for scabs, sores anywhere on the body.

B. Record assessment

a. Breeding records
    i. Good does conceive easily and produce 6 – 8 litters each year and about 15 litters during their life
    ii. Average litter size should be seven or more with few losses of young
    iii. The young should grow rapidly, reaching 2.2 Kg at 8 weeks.

b. Cost and quantity of the feed eaten or the feed conversion ratio

c. Look at the family history of the rabbits for ancestors with good performance records.

d. Performance records
    i. Average litter size should be 8 or more
    ii. Mortality rate < 5%
    iii. Conception rate >90%
    iv. Weight of litters at 4 weeks (total 4.5Kg)
    v. Dressing percentage (55 – 60% including heart, liver and kidneys)
    vi. Feed requirement to produce 1.8 Kg rabbit <6.8Kg.

Equipment used in rabbitry

1. Watering system: It supplies fresh water to rabbit manually using crocks and cans or by automatic system. Automatic systems is the most efficient system. It must be cleaned and periodically checked for leaks and blockage to ensure availability of water. It allows water to be available for 24 hours a day with little or no maintenance.
2. Feeders: feeders are placed high enough to minimize contamination and fastened to prevent tipping over. Feeder trough attached to the cages from the outside are the most common type. Metal is a logical choice than wood which can be chewed on by rabbits. Placing through outside the cage make refilling faster and easier.

3. Nest boxes: Nest boxes are used to house rabbit kits. They should provide seclusion, adequate ventilation and protect litter from drafts. A nest box measuring 30.5 x 30.5 x 61 centimeters with one side cut down to 15.24 cm should be insulated and filled with straw.
4. Cleaning and disinfecting equipment
5. Caring cages
6. Scale
7. Tatoo kit
8. Shovels and wheelbarrow
9. Table or workbench
10. Assorted hand tools for maintenance and repair
11. Nail clippers

Housing and equipment for pigeon production

Pigeons are easily trained to recognize “home.” The wing feathers are clipped and the birds are fed close to the dovecote; by the time they refledge, their homing instinct has been developed. Alternatively, newly captured pigeons may be trained by confining them to the dovecote for at least one week. At first, a little grain is provided in the morning (this is to ensure the birds will return to the coop). The birds can obtain the rest themselves.

Any waterproof house that is easy to clean is suitable for keeping pigeons. Many traditional dovecotes are built of earthenware pots. In Asia and Europe, wooden pigeon towers are generally used.

Unlike chickens, pigeons do not prefer communal roosts. Instead, they prefer nesting shelves, of which there should be two for each breeding pair. The shelves are usually placed in dark corners and are fitted with low walls to keep eggs from rolling out.

Housing and equipment for quail production

It is necessary to keep quail in battery cages on wire floors because males secrete a sticky foam (from the foam gland) with their feces; on a solid floor, this adheres to the feet and collects dung, leading to crippling and breakage of eggs.

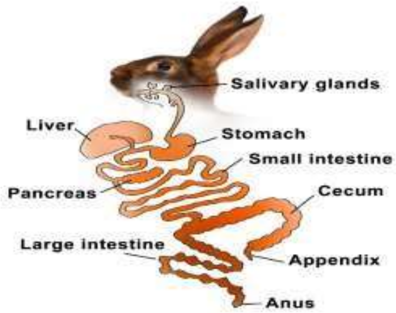
Adult quail can live and produce successfully if they are allowed 80 cm^2 of floor space per bird. However, for reproduction about twice that is needed to allow for mating rituals. If properly mated, high fertility rates and good egg hatchability can be expected. To obtain fertile eggs, one male is needed for roughly six (6) females.

Eggs hatch in about 17 days. Chicks require careful attention. Brooding temperatures of between 31°C and 35°C are needed for the first week and above 21°C for the second week. From the second week on, chicks can survive at room temperature. (These temperatures are similar to those required for common chickens.) In cold climates, supplemental heat may be needed as well as protection from cool drafts.

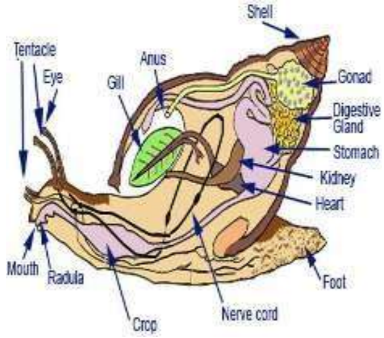
Clean water must be provided at all times, with care taken to prevent the chicks from drowning in their water troughs. Shallow trays, jar lids, or pans filled with marbles or stones may be used.

Some Micro Livestock Digestive System

DIGESTIVE SYSTEM OF A RABBIT



DIGESTIVE SYSTEM OF A SNAIL



A QUAIL BIRD

A GUINEA FOWL

A TYPICAL RABBIT

A TYPICAL SNAIL

FEEDING RESOURCES USED FOR MICRO LIVESTOCK

1. Green Forages and Browse Plants

These are fresh, leafy plant materials fed directly to micro-livestock.

- **Examples:**
  - *Rabbits/Guinea pigs/Grasscutters:*
    - Elephant grass (*Pennisetum purpureum*)
    - Guinea grass (*Panicum maximum*)
    - Sweet potato leaves (*Ipomoea batatas*)
    - Cassava leaves (detoxified)
    - Tridax (*Tridax procumbens*)
    - Centrosema, Stylosanthes, and other legumes
- **Benefits:** Rich in fiber, vitamins, and minerals; improves gut health.

2. Agro-Industrial By-products

These are leftovers from food processing industries that can be repurposed as animal feed.

- **Examples:**
  - **Cassava peels**
  - **Wheat offal/bran**
  - **Maize bran**
  - **Soybean meal**
  - **Palm kernel cake**
  - **Brewers' dried grains (BDG)**
- **Common Users:** Rabbits, quails, grasscutters, guinea pigs

3. Grains and Energy-Rich Concentrates

Used to provide energy for growth and production.

- **Examples:**
  - Maize (cracked or ground)
  - Sorghum
  - Millet
  - Rice bran
  - Groundnut cake (protein-rich too)
- **Common Users:** Quails, pigeons, rabbits (in formulated feeds)

4. Protein-Rich Feedstuffs

Essential for muscle development, reproduction, and immunity.

- **Examples:**
  - Fishmeal
  - Soybean meal
  - Blood meal
  - Termites and maggots (especially for birds like quails and pigeons)
  - Earthworms (used in snail and rabbit production)
- **Common Users:** Quails, pigeons, rabbits, grasscutters

5. Fruits and Vegetables

Serve as supplementary feeds, especially for hydration and vitamins.

- **Examples:**
  - Watermelon
  - Pawpaw (papaya)
  - Carrot
  - Pumpkin
  - Cabbage
  - Cucumber
  - Banana peels
  - Plantain peels

- **Common Users:** Rabbits, guinea pigs, grasscutters, snails

6. Kitchen and Farm Wastes (Non-toxic)

Low-cost option, especially in backyard production.

- **Examples:**
    - Yam peels
    - Plantain and banana peels
    - Vegetable scraps
    - Leftover rice or maize (cooked)
- **Note:** Should be free from salt, oil, or harmful residues.

7. Animal Protein Sources

Especially important for omnivorous or carnivorous micro-livestock.

- **Examples:**
    - Maggots (black soldier fly larvae)
    - Termites
    - Earthworms
    - Crushed snails/shellfish for calcium
- **Common Users:** Quails, pigeons, grasscutters (occasionally)

8. Formulated Pelleted Feeds

These are commercially or home-formulated rations containing a balance of protein, energy, fiber, vitamins, and minerals.

- **Species-Specific Pellets:**
    - Rabbit pellets
    - Quail grower and layer mash
    - Grasscutter formulated feeds
    - Pigeon mix (grains and seeds)
- **Benefits:** Balanced nutrition, ease of handling and storage.

9. Mineral and Vitamin Supplements

Needed for bone health, metabolism, reproduction, and disease resistance.

- **Examples:**
    - Salt licks
    - Bone meal
    - Oyster shell (calcium source)
    - Commercial mineral-vitamin premixes
- **Common Users:** All species, especially reproductive and growing stock

10. Water

Although not a feed per se, **clean, fresh water** is a vital nutrient for all micro-livestock. It aids digestion, thermoregulation, and waste excretion.

Summary Table of Feeding Resources per Micro-Livestock Type

Feed Type	Rabbits	Quails	Grasscutters	Snails	Guinea Pigs	Pigeons
Green forages	✅	🚫	✅	✅	✅	🚫
Grains/Concentrates	✅	✅	✅	🚫	🚫	✅
Agro-industrial byproducts	✅	✅	✅	🚫	✅	✅
Fruits & vegetables	✅	🚫	✅	✅	✅	🚫
Animal protein (insects)	🚫	✅	✅ (limited)	✅	🚫	✅
Pelleted/formulated feeds	✅	✅	✅	🚫	✅	✅
Minerals and vitamins	✅	✅	✅	✅	✅	✅
Water	✅	✅	✅	✅	✅	✅

Nutrient Requirements and Nutritional Disorders in Micro-Livestock

1. Nutrient Requirements of Various Micro-Livestock Species

Micro-livestock require the six essential classes of nutrients for optimal performance:

Essential Nutrients:

- **Carbohydrates:** Provide energy
- **Proteins:** Build body tissues and support growth
- **Fats:** Energy reserve and fat-soluble vitamin absorption
- **Vitamins:** Regulate body processes (e.g., Vitamin A, C, D, E)
- **Minerals:** For bone development, egg shell, blood, etc. (e.g., calcium, phosphorus)
- **Water:** Most critical nutrient; supports digestion, circulation, and excretion

Species-Specific Nutrient Needs:

Species	Nutrient Requirements
Rabbits	High fiber (roughage), moderate protein, low fat, calcium, vitamins A & D
Grasscutters	High fiber diet, moderate protein, rich in calcium, phosphorus, and energy

Species	Nutrient Requirements
Snails	High calcium (for shell formation), low protein, fresh greens, and high humidity levels

Quails	High protein (20–24%), energy-rich grains, calcium, vitamin D3 for egg production
Guinea pigs	Fiber-rich feed, **require dietary Vitamin C**, moderate protein

2. Daily Feed and Water Intake per Species

Species	**Daily Feed Intake**	**Daily Water Intake**
Rabbits	100–150 g dry feed	200–500 ml
Grasscutters	500–800 g fresh forage	400–600 ml
Snails	20–30 g fresh vegetables	2–5 ml (or from environment)
Quails	20–30 g per bird	50–100 ml
Guinea pigs	80–100 g fresh feed or pellets	100–250 ml

Water needs increase during hot weather, reproduction, and lactation.

3. Symptoms of Nutritional Diseases and Disorders in Micro-Livestock

When any of the essential nutrients are lacking, micro-livestock can suffer from nutritional diseases or metabolic disorders. These impact growth, reproduction, and survival.

Common Nutritional Disorders and Their Symptoms

Deficiency	**Symptoms**	**Species Affected**
Protein Deficiency	Poor growth, muscle wasting, rough coat/fur, low reproduction	Rabbits, grasscutters, quails
Vitamin A Deficiency	Eye problems, weak immune response, dry skin/fur	Rabbits, quails
Vitamin C Deficiency	Swollen joints, bleeding gums, poor wound healing, weakness	Guinea pigs (cannot synthesize Vit C)
Calcium Deficiency	Soft/brittle bones, weak shells (snails/quails), poor laying	Snails, quails, rabbits

Deficiency	Symptoms	Species Affected
Phosphorus Deficiency	Weak muscles, poor growth, stiff joints	Grasscutters, guinea pigs
Energy Deficiency	General weakness, stunted growth, poor fertility	All species
Overfeeding (Obesity)	Inactivity, fatty liver, low fertility, joint strain	Rabbits, guinea pigs
Dehydration	Dry mouth, sunken eyes, low feed intake, slow movement	All species

4. Prevention and Control of Nutritional Disorders

- Provide **balanced diets** using species-appropriate feed formulations.
- Always ensure **clean water** is available.
- Supplement with **vitamin-mineral premixes**, especially during stress or disease.
- Feed **fresh forages** rich in fiber and minerals.
- Monitor animal behavior and physical appearance regularly for early signs of deficiency.

Understanding the Reproductivity Micro-Livestock Production

Reproduction is the biological process by which animals produce offspring to ensure the survival and continuity of their species. Good reproductive performance ensures increased production of meat, eggs, and young animals for replacement or sale.

Key Reproductive Terms:

- **Sexual maturity** – Age at which the animal is capable of reproducing
- **Gestation period** – Time between mating and birth
- **Incubation period** – Time it takes for fertilized eggs to hatch (in birds)
- **Litter size** – Number of offspring produced at birth

2. **Mating system** – The pattern of pairing between males and femalesReproductive Characteristics of Common Micro-Livestock

Species	Sexual Maturity	Gestation/Incubation Period	Litter/Clutch Size	Reproductive Features
Rabbits	4–5 months	28–32 days	4–12 kits	Frequent breeders; induced ovulation
Grasscutters	6–9 months	140–155 days	2–6 pups	Polygamous; seasonal breeders
Snails	6–12 months	21–30 days (egg hatching)	50–200 eggs	Hermaphroditic (both sexes in one snail)
Guinea pigs	3–4 months	59–72 days	2–6 pups	Year-round breeders
Quails	6–8 weeks	16–18 days (egg incubation)	6–12 eggs per clutch	High egg producers; need light stimulation

3. Mating Procedures in Micro-Livestock

Each species of micro-livestock has a unique mating process that must be carefully managed for successful breeding.

A. Rabbits

- **Mating System**: Controlled (1 buck to 5–10 does)
- **Procedure**:
  - Always take the **doe to the buck's cage**.
  - Allow the buck to mate; fall-off indicates successful mating.
  - Remove the doe after mating to avoid aggression.
- **Note**: Rabbits can breed again 1–2 weeks after kindling (giving birth).

B. Grasscutters (Cane Rats)

- **Mating System**: Colony system (1 male to 4–6 females)
- **Procedure**:
  - Form breeding colonies with mature animals.
  - Allow natural mating within the group.
  - Remove unproductive or aggressive males.
- **Note**: Pregnancy signs include swollen abdomen and weight gain.

C. Snails

- **Mating System**: Hermaphroditic (each snail has male and female organs)
- **Procedure**:
  - Keep mature snails (6 months and above) in a moist, dark environment.
  - Snails exchange sperm and both partners can lay eggs.
  - Eggs are laid in soft soil and hatch in 3–4 weeks.
- **Note**: Snails reproduce better in the rainy season.

D. Guinea Pigs

- **Mating System**: Polygamous (1 male to 2–3 females)
- **Procedure**:
  - Introduce the male into the female enclosure.
  - Natural mating occurs quietly and mostly at night.
  - Pregnant females should be separated to avoid stress.
- **Note**: Females should not have their first pregnancy after 8 months of age to avoid birthing problems.

E. Quails

- **Mating System**: Polygamous (1 male to 3–5 females)
- **Procedure**:
  - Maintain mating groups in breeder cages.
  - Natural mating occurs as males mount the females.
  - Fertilized eggs are collected daily and incubated artificially or naturally.
- **Note**: Fertility is influenced by light (14–16 hours/day improves egg production).

4. Importance of Proper Mating Management

- Ensures regular and high-quality offspring production
- Prevents inbreeding and infertility
- Helps in selecting genetically superior breeding stock
- Enhances productivity and farm profitability

Conclusion

A sound understanding of the reproductive behavior and mating procedures of micro-livestock is key to successful production. Each species has specific mating strategies, and careful management ensures high productivity, good health, and sustainable operations.

Quick Revision Questions

1. At what age do rabbits reach sexual maturity?
2. Why must the doe be taken to the buck's cage during mating?
3. What type of reproductive system do snails have?
4. State the average gestation period of grasscutters, rabbits and quail.
5. Why should guinea pig females not breed for the first time after 8 months of age?